Interpretable Multi-Scale Household Energy Forecasting via Holt–Winters Feature Augmentation, XGBoost, and Post-hoc Explanations

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***Abstract*—**

Most energy forecasting papers hand you a table with one RMSE per scale and call it done. That number does not tell you when the model breaks down or which features actually matter versus which just look important on a SHAP plot. This paper builds a forecasting framework around that gap. The approach combines Holt–Winters exponential smoothing with XGBoost—an established pattern in the literature—but extends it with leakage-aware feature engineering, temporal error decomposition by hour-of-day, day-of-week, and month, and controlled feature intervention studies through drop-one ablation and lag add-feature experiments. The UCI Individual Household Electric Power Consumption dataset is used for evaluation across hourly, daily, weekly, monthly, and quarterly resolutions. Results show that aggregate RMSE hides non-uniform degradation patterns only visible when error is sliced by time. Adding autoregressive lag features materially improved performance at hourly granularity and shifted SHAP attribution rankings—illustrating that feature importance cannot be interpreted in isolation from what features are present.

***Index Terms*—** Energy forecasting, XGBoost, Holt–Winters, explainable AI, SHAP, LIME, error analysis, feature ablation, time-series cross-validation.

I. Introduction

Residential electricity forecasting sits at the center of smart-grid orchestration, demand response, and household energy optimization. Get it wrong and operating costs rise, scheduling quality drops, and users stop trusting the automated systems built on top of the forecasts. Get it right on the aggregate metric but miss the temporal failure patterns, and the problem is roughly the same—you just do not know it yet.

Gradient-boosted tree models, XGBoost in particular, consistently achieve strong predictive performance on tabular time-series problems. The trade-off is interpretability. A model that a domain user or system operator cannot interrogate is difficult to trust, regardless of its error metrics. Post-hoc explanation methods—SHAP, LIME, permutation importance, partial dependence plots—partially address this, but they report attributions given a fixed feature set. They cannot tell you whether a highly attributed feature is load-bearing or whether removing it would barely move the needle.

The pipeline built in this work starts from an existing approach—Holt–Winters decomposition for feature augmentation fed into XGBoost—and extends it in three concrete directions the base formulation does not cover. First, leakage in the HW component features is fixed by lagging them one step, so that predicting y at $t+1$ uses only values known at t. Second, the RMSE is broken into time slices by hour-of-day, day-of-week, and month, exposing failure windows a single number cannot reveal. Third, feature interventions—drop-one ablation and lag add-feature experiments—measure accuracy changes directly rather than relying on attribution alone.

A. Contributions

- A reproducible HW+XGB+XAI pipeline for household energy forecasting across hourly, daily, weekly, monthly, and quarterly resolutions, runnable end-to-end from a single CLI command.
- Leakage-aware temporal modeling via chronological splits, `TimeSeriesSplit` cross-validation, and one-step-lagged Holt–Winters component features.
- Error decomposition by hour-of-day (24 buckets), day-of-week (7 buckets), and month (12 buckets), exposing non-uniform degradation patterns hidden by aggregate RMSE.
- Feature intervention analysis via drop-one ablation and target-lag add-feature experiments, quantifying accuracy deltas and corresponding SHAP attribution shifts.
- Advanced XAI extensions: SHAP interaction effects, counterfactual recourse recommendations for target reduction, and a confidence-vs- explanation stability analysis linking conformal interval width with LIME consistency.

II. Literature Review

A. Household Energy Forecasting

Household electricity forecasting methods split into three broad groups: classical statistical models such as exponential smoothing and ARIMA variants, machine learning approaches built around tree ensembles, and deep learning architectures [5]. Statistical models handle seasonality well but underfit non-linear demand patterns. Deep learning models capture complex dependencies but require substantial data and are difficult to explain to system operators. Tree ensembles sit in the middle: strong

predictive accuracy, moderate training cost, and amenable to post-hoc explanation [2].

Hybrid approaches combining classical decomposition with machine learning are an active area. Holt–Winters exponential smoothing is a common decomposition choice because it handles both trend and seasonality with few parameters and is computationally cheap. Most published HW+ML pipelines either forecast each component separately and combine outputs, or feed component estimates as features to a downstream learner. This work takes the latter approach.

B. Interpretable Machine Learning for Energy Systems

Post-hoc interpretability methods divide into global and local explanations. SHAP provides additive feature attributions grounded in cooperative game theory, with efficient TreeSHAP implementations for gradient-boosted models [3]. LIME fits a local linear surrogate around a specific prediction [4]. Permutation Feature Importance (PFI) shuffles one feature at a time and measures the accuracy drop. Partial Dependence Plots (PDP) show the marginal effect of a feature averaged over the distribution of other inputs [8].

Using multiple methods together is more informative than any single one. SHAP and PFI do not always agree, and that disagreement often indicates feature interactions or instability under perturbation—both worth reporting rather than explaining away.

C. Research Gap

The standard evaluation practice in energy forecasting is to report a single aggregate metric per scale and present SHAP or PFI plots as evidence of interpretability. This has two problems. First, aggregate RMSE obscures temporal failure concentration. Second, feature attribution without feature intervention does not establish causal importance—a feature can rank highly on SHAP while being almost entirely redundant. This work addresses both gaps directly.

III. Dataset and Preprocessing

A. Dataset

All experiments use the UCI Individual Household Electric Power Consumption dataset [1], recorded at one-minute resolution over roughly four years from a single household in Sceaux, France. The dataset includes global active power P_{global}, reactive power Q_{global}, voltage V, global intensity I, and three sub-metering channels S_1, S_2, S_3 corresponding to the kitchen, laundry room, and water heater or air conditioning loads.

B. Cleaning and Feature Construction

Let $\mathcal{T} = \{t_1, t_2, \ldots, t_N\}$ denote the full timestamp index. Missing values are filled using time-based linear interpolation, where t_a and t_b are the nearest non-missing timestamps bracketing t, followed by forward and backward fill for residual gaps.

Cyclical time encodings are added to avoid boundary discontinuities with raw integer features. For hour-of-day $h \in \{0, \ldots, 23\}$ and day-of-week $d \in \{0, \ldots, 6\}$: using sine/cosine transformations for each cycle. These encodings give the model a smooth periodic representation of time: Euclidean distance between any two time points is proportional to their circular distance on the clock or week cycle.

C. Multi-Scale Aggregation

The one-minute series is resampled to five temporal granularities $\mathcal{G} = \{\text{hourly, daily, weekly, monthly, quarterly}\}$. For a time window $\mathcal{W}$ of width $g \in \mathcal{G}$: we use mean aggregation for power-like variables and sum aggregation for sub-metering channels. Summing the energy channels preserves total energy consumed per period. A stationarity check via the Augmented Dickey–Fuller (ADF) test is run on the training target per scale, where $\hat{\delta}$ is the OLS estimate of the unit-root coefficient in $\Delta y_t = \alpha + \beta t + \delta y_{t-1} + \sum_{k=1}^{p} \gamma_k \Delta y_{t-k} + \varepsilon_t$. Results are written to JSON as a diagnostic only.

IV. Methodology

A. Holt–Winters Feature Augmentation

The additive Holt–Winters model decomposes the training target y_t:

$$y_t = \ell_t + b_t + s_t + \varepsilon_t, \tag{1}$$

where ℓ_t is the level, b_t the trend, s_t the seasonal component with period m, and ε_t the residual.

The standard HW+ML formulation suffers from leakage: component estimates at t depend on y_t yet are used to predict y_t. The fix applied here lags all components by one step:

$$\boldsymbol{\phi}_t = \left[\hat{\ell}_{t-1},\ \hat{b}_{t-1},\ \hat{s}_{t-1},\ \hat{\varepsilon}_{t-1}\right]^\top, \tag{2}$$

so that predicting y_{t+1} uses only component values observable at t.

B. Supervised Learning Formulation

The feature vector at time t combines original covariates, cyclical time encodings, lagged HW components from (2), and an optional autoregressive lag block (e.g., $y_{t-1}, y_{t-24}, y_{t-168}$). The prediction target is y_{t+1}:

$$\hat{y}_{t+1} = f(\mathbf{x}_t). \tag{3}$$

C. XGBoost Regression and Tuning

XGBoost learns an additive ensemble of K regression trees:

$$\hat{y}_{t+1} = \sum_{k=1}^{K} f_k(\mathbf{x}_t), \quad f_k \in \mathcal{F}. \tag{4}$$

The objective at iteration k is: with a squared-error loss and tree-complexity regularisation. where T is the number of leaves, $\mathbf{w}$ the leaf weight vector, γ the minimum split gain, and λ the L_2 penalty. Hyperparameter tuning uses `TimeSeriesSplit` (n=5) and `GridSearchCV`. Feature standardisation: is applied using training-set statistics only, with $\mu_j^{\text{train}}, \sigma_j^{\text{train}}$ computed on training data only.

V. Explainability and Diagnostics

A. Post-hoc Explanation Methods

Four complementary methods are applied to each trained model.

SHAP (TreeSHAP) computes Shapley-value attributions for each feature, where $v(S)$ is the model output conditioned on features in S [3]. TreeSHAP runs in $\mathcal{O}(TLD^2)$ time.

LIME fits a local linear surrogate g around instance $\mathbf{x}_0$ by minimizing weighted local approximation error, with proximity kernel $\pi_{\mathbf{x}_0}(z) = \exp\big(-D(z, \mathbf{x}_0)^2/\sigma^2\big)$ [4].

Permutation Feature Importance (PFI) measures error increase after shuffling one feature at a time, where $\mathbf{X}_{\pi_j}$ has column j randomly permuted.

Partial Dependence (PDP) estimates the marginal effect of feature j by averaging predictions over the empirical distribution of all other features.

SHAP interaction dependence is used to visualize pairwise effects by plotting a feature's SHAP value while color-coding by an interacting feature. This reveals conditional behavior hidden by single-feature rankings.

Counterfactual recourse addresses the practical question: "what should change to reduce predicted consumption by 20%?" We search plausible feature ranges and report minimal-change suggestions that move predictions toward that target.

B. Error Decomposition by Time Slice

For each temporal bucket b in {hour-of-day, day-of-week, month}, the slice RMSE is:

$$\text{RMSE}_b = \sqrt{\frac{1}{|\mathcal{I}_b|} \sum_{t \in \mathcal{I}_b} \big(y_{t+1} - \hat{y}_{t+1}\big)^2}, \tag{5}$$

where $\mathcal{I}_b = \{t \in \mathcal{T}_{\text{test}} : \text{bucket}(t) = b\}$. The 24 hourly, 7 daily, and 12 monthly slices together produce a map of where the model degrades—invisible in the scalar aggregate.

C. Feature Impact Studies

Drop-one ablation. For each feature j in the top-N by XGBoost gain, retrain on $\mathcal{F} \setminus \{j\}$ and compute:

$$\Delta\text{RMSE}_j = \text{RMSE}(f_{\mathcal{F}\setminus\{j\}}) - \text{RMSE}(f_{\mathcal{F}}), \tag{6}$$

$$\Delta R^2_j = R^2(f_{\mathcal{F}}) - R^2\big(f_{\mathcal{F}\setminus\{j\}}\big)\,. \tag{7}$$

Positive ΔRMSE_j indicates the feature was genuinely load-bearing; zero or negative values suggest redundancy.

Add-feature experiment. Let $\mathcal{F}^+ = \mathcal{F} \cup \mathbf{l}_t$ be the augmented feature set. Record:

$$\Delta\text{RMSE}^+ = \text{RMSE}(f_{\mathcal{F}}) - \text{RMSE}(f_{\mathcal{F}^+}), \tag{8}$$

alongside SHAP ranking shifts between $f_{\mathcal{F}}$ and $f_{\mathcal{F}^+}$.

D. Uncertainty and Explanation Stability

Split conformal prediction intervals are computed at 90% and 95% levels. With calibration residual quantile $\hat{q}_{1-\alpha}$, the interval for each test point is $[\hat{y}_{t+1} - \hat{q}_{1-\alpha}, \hat{y}_{t+1} + \hat{q}_{1-\alpha}]$. We report empirical coverage and average width.

To connect uncertainty with explanation reliability, LIME is rerun with multiple seeds per instance. Stability is measured as mean pairwise cosine similarity of normalized local weight vectors, then correlated (Spearman) with 95% interval width.

VI. Experimental Setup

A. Evaluation Metrics

Four metrics are reported for test set $\{(y_i, \hat{y}_i)\}_{i=1}^n$:

$$\text{MAE} = \tfrac{1}{n}\sum|y_i - \hat{y}_i|, \tag{9}$$

$$\text{MSE} = \tfrac{1}{n}\sum(y_i - \hat{y}_i)^2, \tag{10}$$

$$\text{RMSE} = \sqrt{\tfrac{1}{n}\sum(y_i - \hat{y}_i)^2}, \tag{11}$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2}. \tag{12}$$

RMSE is the primary comparison metric. R^2 provides a scale-independent reference. MAE is reported as a robustness check.

B. Train/Test Protocol

An 80/20 chronological split is used throughout; no shuffling occurs at any stage. Hyperparameter selection uses five-fold `TimeSeriesSplit` cross-validation applied exclusively to the training portion. The test set is touched once only, after hyperparameters are selected, to compute final metrics.

VII. Results

A. Aggregate Performance Across Scales

Table 1 reports MAE, RMSE, and R^2 for each temporal scale. Performance generally improves at coarser scales—aggregation smooths short-term variability that is difficult to predict from available covariates alone. The hourly scale is the hardest, with the most volatile target and the strongest dependency on recent history.

The negative R^2 at daily scale (-0.2280) is a clear signal: the model underperforms a naive mean predictor at that granularity. This likely reflects the smoothing effect of daily aggregation reducing the variance the model can exploit, combined with insufficient feature information to capture day-to-day fluctuations. Monthly ($R^2 = 0.7580$) and quarterly ($R^2 = 0.8127$) scales show substantially stronger fit, where seasonal and trend components dominate and are well-captured by the HW augmentation.

B. Temporal Error Decomposition

Slicing test RMSE by hour-of-day, day-of-week, and month using (5) confirms that aggregate metrics hide operationally important variation. Evening hours, when household demand peaks and usage patterns are more variable, consistently show elevated RMSE relative to overnight windows. Day-of-week slices reveal higher error on weekdays at certain scales, reflecting work-schedule-dependent consumption patterns the model captures imperfectly. Monthly slices expose seasonal variation: winter months show higher absolute error where heating or cooling loads dominate. These patterns indicate where confidence intervals should be widened and where additional features—weather, occupancy proxies—would provide the most benefit.

TABLE 1. Forecasting Performance by Temporal Scale

Scale	MAE	RMSE	R^2
Hourly	0.4002	0.5258	0.4787
Daily	0.2842	0.3457	−0.2280
Weekly	0.1432	0.1977	0.3356
Monthly	0.0902	0.1037	0.7580
Quarterly	0.0627	0.0721	0.8127

Source: `outputs/tables/metrics_latest.csv`. Negative R^2 at daily scale indicates the model underperforms a naive mean predictor at that granularity.

C. Feature Ablation and Lag Add-Feature Results

Drop-one ablation, computed via (6)–(7), shows that not all top-SHAP features are equally load-bearing. Several features that ranked highly by gain showed only modest ΔRMSE when removed, suggesting shared predictive contribution with correlated features. A smaller subset showed clear positive ΔRMSE on removal, confirming genuine predictive necessity.

Adding target lag features at hourly resolution improved RMSE materially, with ΔRMSE^+ from (8) driven primarily by the 1-step and 24-step lags. The SHAP ranking shifted after lags were added: HW-derived features dropped in relative attribution because the autoregressive lags absorbed much of the short-term predictive load. This demonstrates why feature importance rankings must not be interpreted independently of what features are present.

D. Explainability Outcomes

SHAP summary plots show global feature ranking and directionality. The HW level component $\hat{\ell}_{t-1}$ consistently ranks among the top features—level tracks the current baseline of consumption, and the next-step value is closely tied to it. Cyclical time features (hour_sin, hour_cos) rank highly at hourly scale and less so at coarser scales where within-day variation is averaged out.

LIME explanations for selected test instances show that local feature rankings can differ substantially from global SHAP rankings, particularly for unusual consumption periods. PFI rankings do not always match SHAP rankings—a signal of features whose importance depends on interaction effects rather than independent marginal contribution. PDP plots confirm expected monotonic relationships for level and trend components and nonlinear response curves for cyclical time features.

SHAP interaction plots reveal that feature influence is context-dependent. For example, the importance of appliance submetering can increase in specific time windows rather than acting uniformly over the day.

Counterfactual outputs provide actionable recourse by suggesting feature changes associated with lower predicted consumption. These are model-based recommendations (not causal guarantees), but they improve practical interpretability for end users.

Confidence-vs-stability analysis shows that instances with wider 95% intervals tend to have less stable LIME explanations, indicating local explanations are less reliable when model uncertainty is high.

VIII. DISCUSSION

Three practical points stand out.

First, aggregate metrics are not enough for deployment decisions. A model that looks fine on average can be substantially degraded during the hours and seasons that matter most. Temporal error decomposition (5) is cheap to compute and adds real operational value—it should be a default part of any evaluation pipeline, not an optional diagnostic.

Second, feature attribution is not the same as feature necessity. High SHAP values are often shared across correlated features. Ablation studies (6)–(7) are the more direct test, and their results do not always match attribution rankings. Both should be reported when interpretability matters for deployment.

Third, the HW+XGB combination offers a reasonable middle ground between fully statistical and fully black-box approaches. The Holt–Winters decomposition are transparent and produce interpretable components. XGBoost handles the non-linear residuals via (4). The combination is not the highest-accuracy possible approach, but it is substantially more interpretable and easier to audit.

Fourth, combining uncertainty and explanation stability is more informative than either alone. A local explanation attached to a high-uncertainty prediction should be treated as exploratory rather than definitive.

A. Threats to Validity

The main limitations are single-dataset evaluation and limited hyperparameter search breadth. The dataset covers a single household in France; generalisation to different household types, climates, or grid contexts is an open question. The daily-scale negative R^2 warrants further investigation: it may reflect aggregation collapsing the predictive signal present at finer granularities.

IX. CONCLUSION AND FUTURE WORK

This paper built and evaluated an interpretable multi-scale energy forecasting pipeline combining Holt–Winters feature augmentation (1)–(2), XGBoost regression (4), and four post-hoc explanation methods. The core prediction approach existed in prior work. What was added— leakage-aware component lagging, temporal error decomposition, drop-one ablation, and lag add-feature experiments—addresses the evaluation and interpretability gaps most published HW+ML work leaves open.

The main result is not a new state-of-the-art accuracy number. It is a more honest picture of where a well-built baseline model succeeds and where it struggles, and a cleaner methodology for connecting feature importance claims to actual predictive necessity. The added SHAP interaction, counterfactual, and confidence-vs-stability analyses turn the interpretability layer into a stronger decision-support workflow. The daily-scale result ($R^2 = -0.228$) is a useful reminder that aggregation does not always help: smoothing can collapse the very signal a model

needs.

Future work will focus on: (i) cross-dataset generalisation to evaluate robustness across household types and geographies; (ii) causal and feasibility-constrained counterfactuals; (iii) concept-drift detection and adaptive retraining; and (iv) richer contextual features—weather data, occupancy proxies—most likely to reduce error in the peak-period windows identified by the error decomposition analysis.

REPRODUCIBILITY

All experiments run end-to-end from a single CLI entry point. Outputs are organised under `outputs/` as follows: `metrics_latest.csv` and `metrics_history.csv` for performance; `error_slice_*` tables and plots for temporal decomposition; `ablation_drop1_*` for ablation; `pfi_*` for permutation importance; `shap_summary_*` and `pdp_*` for global explanations; `shap_interactions_*` for interaction effects; `counterfactuals_*` for recourse recommendations; `intervals_*` for uncertainty bands; and `lime_*` plus `lime_stability_*` for local explanations and stability analysis.

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